

Samuel Chiu PIFSC GitHub Workshop 2026 Lesson Plan (Demos)

Getting Started:

- Make sure everyone has Prerequisites installed:
 - GitHub
 - R-Studio
- GitHub Account
- Repositories Folder

Repo Creation & GitHub Exploration

- Show People Github and explore
 - Searching for code
 - Downloading random repositories
 - GitHub Features
 - Projects
 - Starred projects
 - Following
 - Issues
- Create first repo for Profile
 - Public vs Private
 - change privacy
 - Always private start
 - Generate README with Gemini
 - Make Public
- Create a Second Repo For Testing
 - Clone the repo to our device.
 - Shown in both ways
 - CLI and R Studio
- Add an existing project to a GitHub Repo
 - Dragging files over
 - Adding the Remote URL

Committing and Pushing Demo

Demonstrate how to make a commit and push with R studio

- Navigate to the Git pane.
- Add a File
- Commit said File

- Super informative Message

- Push the File
- Check the file on github

Demonstrate this process with CLI

Branching Demo

Show R-Studio Branching

- Create branch
- Swap branches
- Make People understand the diagram

Show CLI Branching

- Make a new branch
- Switch to branch
 - Set Upstream

Show Branching Diagram

- R Studio
- VS Code

Pulling Demo

- Pull onto branch
- Github Pull Request

Merge Conflict Demo

- Create a Merge Conflict
- Show merge conflict
 - R Studio
 - VS Code
- Resolve Merge Conflict

Security Demo

Create and Modify .gitignore\

Advanced Demo

- Github Pages
- Github Actions
 - PIFSC Publications Action

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Extra Material (Not Lesson plan but helps)

Copy and paste:

```
<<<<<< HEAD
# This is the code currently on your active
branch
data <- read_csv("data/raw_data_v2.csv")
=====
# This is the code on the branch you are
trying to pull/merge in
data <- read_csv("data/cleaned_data.csv")
>>>>>> origin/main
```